

Subham Sharma

+917988944185 | subham1401sh@gmail.com | linkedin.com/in/subham1401

Professional Summary

AI Engineer specializing in Generative AI, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), and FastAPI backend development. Experienced in building production-ready LLM applications, AI agents, and scalable cloud-native AI systems. Passionate about AI infrastructure, workflow automation, and autonomous agent architectures.

WORK EXPERIENCE

Lead AI Engineer - Freelance – [Aceplace.ai by NovaXQuantum](#) | Remote | Jan 2026 - Jul 2026

- Developed a deterministic, identity-based multi-agent execution runtime with role-based workflows and workflow orchestration.
- Designed the Execution Envelope system with Firestore-backed persistence, crash recovery, concurrency control, and audit logging for reliable execution.
- Built a Next.js monitoring workstation with human-in-the-loop approvals for real-time orchestration and execution traceability.

Generative AI Engineer Intern - [Digital Nexus AI](#) | Remote | Jan 2026 - Jul 2026

- Developed Python-based web scraping pipelines and APIs to collect, process, and manage 50,000+ tender records for AI-powered applications.
- Built RAG pipelines using LangChain, FAISS, Pinecone, embeddings, and LLMs, implementing document indexing and contextual retrieval for accurate semantic search and question answering.
- Integrated FastAPI backend services and collaborated on AI workflows, vector database optimization, and LLM-powered features for scalable GenAI solutions.

Data Analytics Intern - [MacroEdtech](#) | Remote | Aug 2025 - Jan 2026

- Performed data preprocessing, EDA, and feature engineering on 15 real-world datasets; implemented and evaluated supervised and unsupervised machine learning models using performance metrics for model selection.
- Developed structured educational content, technical documentation, and curriculum-aligned learning materials to support scalable data science and machine learning workshops.
- Delivered all assigned projects within deadlines, demonstrating strong analytical thinking, model evaluation skills, and professional execution.

[AI Developer](#) - Freelance - [NovaXQuantum](#) | Remote | Jul 2025 - Aug 2025

- Built an Autonomous Mail Reply Agent using SMTP, IMAP, OpenAI GPT, LangChain, and PostgreSQL to automate intelligent email classification and response generation, processing thousands of emails daily.
- Reduced manual triage by 80% and improved response speed by 90% through fault-tolerant AI workflows featuring automated prioritization, retries, rate-limit handling, fallback logic, and immutable JSON audit logging with 99.9% traceability.

PROJECTS

[ExpenseTracker MCP Server – Cloud Hosted](#)

Tech Stack: Python (async), MCP, FastMCP, SQLite, Docker, Render

- Built a production-ready MCP server enabling secure, concurrent tool execution for multiple AI clients using asynchronous Python.
- Implemented **async, non-blocking APIs** to support concurrent multi-client access.
- Integrated **SQLite via FastMCP** for persistent, lightweight storage.
- Deployed as a **public remote MCP service**, consumable by any compatible AI client.

[FastAPI RAG AI Agent – AWS Hosted](#)

Tech Stack: Python, FastAPI, LangChain, OpenAI GPT, Embeddings, FAISS, REST APIs, Docker, AWS EC2, PDF Loaders, Session Memory

- Designed and deployed a cloud-native RAG system using FAISS, OpenAI embeddings, and FastAPI, enabling accurate semantic search over policy PDFs with low-latency responses.
- Architected a **modular, scalable AI agent layer** with session-based conversational memory, exposed via RESTful APIs with validation and structured error handling, supporting reliable multi-turn interactions.
- Containerized with Docker and deployed on AWS EC2, delivering a production-ready, scalable inference pipeline with improved system reliability and maintainability.

TECHNICAL SKILLS

- **Languages:** C, C++, Python, JavaScript

- **AI & Agent Systems:** LangChain, LangGraph, LangSmith, MCP (Server & Client), OpenAI GPT, Gemini, Ollama, Hugging Face, Prompt Engineering, AI Agents
- **Backend & Web:** FastAPI, REST APIs, HTML, CSS, JavaScript, Streamlit, Web Scraping
- **Databases & Vector Stores:** PostgreSQL, MySQL, SQLite, FAISS, Pinecone, Firebase, Supabase
- **Cloud & DevOps:** AWS (EC2), Docker, GitHub Actions (CI/CD), Cloud Deployment
- **ML & Data:** NumPy, Pandas, scikit-learn, Matplotlib, Seaborn
- **Developer Tools:** GitHub, VS Code, Jupyter, Colab, Android Studio
- **Fundamentals:** Data Structures and Algorithms (DSA), Operating Systems, Computer Architecture

EDUCATION

J.C. Bose University of Science and Technology (YMCA)

Bachelor of Technology in Computer Science and Engineering, Expected June 2027

CGPA: 7.77

Additional Details

Portfolio : <https://subham-portfolio-1101.vercel.app>

Github : <https://github.com/subham012005>

Fiverr : <https://www.fiverr.com/subham0017>